

Sol 1 (Discrete)

In 52 Poker playing cards: -

Ace $\rightarrow 4$, Jack $\rightarrow 4$, Queen $\rightarrow 4$, King $\rightarrow 4$
Clubs $\rightarrow 13$ (Black), Diamonds $\rightarrow 13$ (Red)
 $\downarrow$ $\downarrow$
(1J, 1Q, 1K) (1J, 1Q, 1K)

Hearts $\rightarrow 13$ (Red), Spades $\rightarrow 13$ (Black)
 $\downarrow$ $\downarrow$
(1J, 1Q, 1K) (1J, 1Q, 1K)

3 cards are drawn from a deck of 52 cards. A deck has Total 13 Spades

Total cards = 52

No. of Spades = 13

$\Rightarrow$ No. of Non-Spades = $52 - 13 = 39$

3 cards are drawn $\Rightarrow X = \{0, 1, 2, 3\}$

When $X = 0$ (No Spades $\Rightarrow$ all Non Spades)

$$P(X=0) = \frac{{}^{13}C_0 \times {}^{39}C_3}{{}^{52}C_3} \approx \underline{0.249}$$

When $X = 1$ (one Spade & others are Non-Sp.)

$$P(X=1) = \frac{{}^{13}C_1 \times {}^{39}C_2}{{}^{52}C_3} \approx \underline{0.436}$$

Similarly,

$$P(X=2) = \frac{{}^{13}C_2 \times {}^{39}C_1}{52C_3} \approx 0.352$$

$$P(X=3) = \frac{{}^{13}C_3 \times {}^{39}C_0}{52C_3} \approx 0.013$$

X	$P(X)$
0	0.249
1	0.436
2	0.352
3	0.013

Solⁿ (2) $f(x) = c \binom{2}{x} \binom{3}{3-x}; x=0,1,2$

This is nothing but binomial expression

~~$f(x) = c \binom{2}{x}$~~

Take $x=0$ & $\sum f(x) = 1$

You will get value of c

Ans: $\boxed{c=0.1}$

Solⁿ (3) $x: 0 \quad 1 \quad 2 \quad 3 \quad 4$

$f(x): 0.41 \quad 0.37 \quad 0.16 \quad 0.05 \quad 0.01$

Find CDF of X

$$F(x) = P(X \leq x)$$

$$F(0) = P(X \leq 0) = f(0) = 0.41$$

$$F(1) = P(X \leq 1) = \underbrace{f(0)}_{\text{(previous value of } f(x) \equiv f(0))} + f(1)$$

$$= 0.78$$

Similarly,

$$F(2) = f(0) + f(1) + f(2) = 0.94$$

$$F(3) = P(X \leq 3) = 0.99$$

$$F(4) = P(X \leq 4) = 1$$

Solⁿ (4) Already discussed!

Find the mean (expected) value of
 $g(x) = 2x - 1$

$$E[g(x)] = E[2x - 1]$$

$$= 2E(x) - 1 \quad \text{--- (1)}$$

$$E(x) = ?$$

$$E(x) = \sum x p(x)$$

$$= (x_1 \times p_1) + \dots + (x_n \times p_n)$$

Ans: $\boxed{\frac{38}{3}}$

Ques: 5 Already discussed

Find Variance of $g(x) = 2x + 3$

~~$E[g(x)] = E$~~

$$E\{[g(x) - \mu_{g(x)}]^2\} \text{ --- (1)}$$

First find $\mu_{g(x)}$

$$\begin{aligned}\mu_{g(x)} &= E[g(x)] = E[2x + 3] \\ &= 2E(x) + 3 \text{ --- (2)}\end{aligned}$$

Find $E(x) = \sum x f(x)$

~~Find~~

First find $E(x) = 3/2$ (After solving $E(x)$)

Now from Equation (2):

$$\mu_{g(x)} = 2E(x) + 3 = 6$$

Now From (1)

$$\begin{aligned}E\{[(2x+3)-6]^2\} &= E\{[2x-3]^2\} \\ &= E[4x^2 + 9 - 12x] \\ &= 4E(x^2) + 9 - 12E(x) \end{aligned}$$

Find $E(x^2) = \sum x^2 f(x) = \checkmark$

$$= (x_1^2 \times f_1) + \dots + (x_n^2 \times f_n)$$

Ans: 4

solⁿ 6:

x	0	1	2	3
$p(x)$	0.1	0.3	0.5	0.1

$$Y = X^2 + 2X \equiv g(X) \text{ (Let)}$$

at $x_i = 0, 1, 2$

$$y_i = (0^2 + 2 \times 0), (1^2 + 2 \times 1), (2^2 + 2 \times 2), (3^2 + 2 \times 3)$$

$$y_i = 0, 3, 8, 15$$

hint

y	0	3	8	15
$p(y)$	0.1	0.3	0.5	0.1

$$\text{Mean} = 6.4$$

$$\text{Variance} = 16.24$$

Ques 1 (Contine. Q. v)

(a) $\int_0^{\infty} f(x) dx = 1$ (Hint)

(b) $P(0 \leq x \leq 1) = \int_0^1 f(x) dx = (1/4)$

(c) Find $F(x)$ using $P(0 < x \leq 1)$

Note Find $F(x)$ using $P(a < x \leq b)$
 $\Rightarrow F(x) = F(b) - F(a)$

= This will be the Answer

Ques 2 Already discussed.

Hint: $\int_0^a f(x) dx = \int_a^1 f(x) dx$

$$a^3 = 1/2 \Rightarrow a = 0.7937$$

$$\int_b^1 f(x) dx = 0.05$$

$$b^3 = 95$$

$$b = 0.983$$

Ques 3 Already discussed

(a) $a = 1/2$

(b) CDF of $X \rightarrow$

$$F(x) = P(X \leq x)$$

$$\rightarrow F(x) = 0 \quad ; \quad \text{when } x < 0$$

$$\begin{aligned} \rightarrow F(x) = P(X \leq x) &= \int_0^x f(t) dt \\ &= \int_0^x \frac{t}{2} dt = \frac{x^2}{4} \end{aligned}$$

$$F(x) = \frac{x^2}{4} \quad ; \quad 0 \leq x \leq 1$$

$$\begin{aligned} \rightarrow F(x) &= \int_0^1 \frac{x}{2} dx + \int_1^x \frac{1}{2} dx = \left(\frac{x}{2} - \frac{1}{4} \right) \\ &= \left(\frac{x}{2} - \frac{1}{4} \right) \quad ; \quad 1 \leq x \leq 2 \end{aligned}$$

$$\rightarrow F(x) = \left(\frac{3}{2}x - \frac{x^2}{4} - \frac{5}{4} \right) \quad ; \quad 2 \leq x \leq 3$$

$$\rightarrow F(x) = 1 \quad , \quad \text{when } x > 3$$

$$(c) \quad P(X > 1.5) = \int_{1.5}^2 f(x) dx$$

$$= \int_{1.5}^2 \frac{1}{2} dx + \int_2^3 \left(\frac{3}{2} - \frac{x}{2} \right) dx = \left(\frac{1}{2} \right)$$

Saiv(4) Already discussed

$$(i) \quad K = 2/27$$

$$(ii) \quad P(X < 4) = \int_2^4 f(x) dx = \frac{16}{27}$$

Qun 5 $\left[f(x) = \frac{dF(x)}{dx} \right]$

Already discussed

$$f(x) = x e^{-x}$$

$$E(x) = \int_0^{\infty} x [x e^{-x}] dx = 2$$

$$E(x^2) = \int_0^{\infty} x^2 (x e^{-x}) dx = 6$$

$$\Rightarrow V(x) = E(x^2) - [E(x)]^2 = 2$$

Qun 6 $f(x) = \begin{cases} \frac{2(x+2)}{5}, & 0 < x < 1 \\ 0, & \text{else} \end{cases}$

(i) $P(0 < x < 1) = 1$ (Show)

hint: $\int_0^1 f(x) dx$

(ii) Find the prob. that more than $\frac{1}{4}$ but fewer than $\frac{1}{2}$ of the people contacted will respond

$\rightarrow$ That means $P\left(\frac{1}{4} < x < \frac{1}{2}\right)$

$$= \int_{1/4}^{1/2} f(x) dx = 0.237 \text{ (Ans.)}$$

Scanned: $f(x) = \begin{cases} \frac{20000}{(x+100)^3}, & x > 0 \\ 0, & \text{else} \end{cases}$

(Hint)

(i) $P(x > 200)$

~~Ans:~~
 $= \int_{200}^{\infty} f(x) dx = \left(\frac{1}{9} \right)$

(ii) $P(80 \leq x \leq 120)$

$= \int_{80}^{120} f(x) dx = (0.102)$
